# AMAZON MARKETING BUDGET OPTIMIZATION AGENT

AI-Driven Performance Analysis, Predictive Forecasting & Budget Reallocation Report

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## 1. Executive Summary & Financial Highlights

This executive management report evaluates multi-channel advertising performance across Amazon Sponsored Products, Sponsored Brands, Sponsored Display, Google Ads, Meta Ads, Email, and Affiliate channels. Over the analyzed period, total marketing expenditure was **$256,067.91**, generating **$825,336.45** in attributed revenue at a blended ROAS of **3.22x** and **$32,409.68** in estimated marketing contribution across **12,608** customer conversions.

| Total Spend | Attributed Revenue | Blended ROAS | Marketing Contribution | Total Conversions |
|---|---|---|---|---|
| **$256,067.91** | **$825,336.45** | **3.22x** | **$32,409.68** | **12,608** |

## 2. Multi-Channel Advertising Performance Comparison

| Marketing Channel | Spend ($) | Revenue ($) | ROAS | CPA ($) | Contrib ($) | ROI (%) |
|---|---|---|---|---|---|---|
| Amazon Sponsored Products | $83,025.48 | $328,393.64 | 3.96x | $14.65 | $32,956.26 | 39.69% |
| Amazon Sponsored Brands | $42,503.01 | $117,061.57 | 2.75x | $26.66 | $500.74 | 1.18% |
| Google Ads | $41,426.44 | $92,992.81 | 2.24x | $34.81 | $-9,515.34 | -22.97% |
| Amazon Sponsored Display | $38,684.39 | $51,778.53 | 1.34x | $50.04 | $-21,054.70 | -54.43% |
| Meta Ads | $25,035.57 | $26,051.52 | 1.04x | $63.70 | $-15,529.11 | -62.03% |
| Affiliate Marketing | $15,771.33 | $83,809.00 | 5.31x | $14.47 | $10,098.65 | 64.03% |
| Email Marketing | $9,621.69 | $125,249.38 | 13.02x | $5.06 | $34,953.18 | 363.27% |

## 3. Underperforming & Potentially Inefficient Campaigns

Campaigns identified as underperforming or inefficient based on sub-target ROAS, inflated CPA, and negative marketing contribution with sufficient statistical observation.

| Campaign Name | Channel | Spend | ROAS | CPA | Classification Reason |
|---|---|---|---|---|---|
| AirFryer_Pro_Competitor | Amazon Sponsored Products | $14,508.38 | 1.38x | $40.53 | ROAS (1.38x) is significantly below target (2.50x) and CPA ($40.53) exceeds threshold ($35.00), generating negative mark... |
| Activewear_Collection_Spotlight | Amazon Sponsored Brands | $12,622.72 | 0.99x | $64.07 | ROAS (0.99x) is significantly below target (2.50x) and CPA ($64.07) exceeds threshold ($35.00), generating negative mark... |
| Google_Search_Winter_Jackets | Google Ads | $12,200.60 | 0.76x | $85.32 | ROAS (0.76x) is significantly below target (2.50x) and CPA ($85.32) exceeds threshold ($35.00), generating negative mark... |
| CrossSell_KitchenAppliances | Amazon Sponsored Display | $11,383.83 | 0.75x | $84.95 | ROAS (0.75x) is significantly below target (2.50x) and CPA ($84.95) exceeds threshold ($35.00), generating negative mark... |

| | | | | | |
|---|---|---|---|---|---|
| CompetitorASIN_Beauty_Defense | Amazon Sponsored Display | $10,951.94 | 0.54x | $93.61 | ROAS (0.54x) is significantly below target (2.50x) and CPA ($93.61) exceeds threshold ($35.00), generating negative mark... |

## 4. Recommended Marketing Budget Reallocation (SciPy SLSQP)

The AI optimizer evaluated non-linear diminishing marginal returns subject to the total budget of **$256,067.91**, isolating an experiment reserve of **$12,803.40 (5.0%)**. Reallocation produces an estimated incremental marketing contribution of **+$54,968.11**.

| Channel | Current ($) | Recommended ($) | Shift ($) | Shift (%) | Exp. Revenue ($) | Exp. Contrib ($) |
|---|---|---|---|---|---|---|
| Amazon Sponsored Products | $83,025.48 | $120,212.17 | +$37,186.69 | +44.8% | $430,667.06 | $31,813.30 |
| Affiliate Marketing | $15,771.33 | $30,445.02 | +$14,673.69 | +93.0% | $133,630.89 | $10,846.93 |
| Amazon Sponsored Brands | $42,503.01 | $29,381.03 | $-13,121.98 | -30.9% | $87,513.27 | $2,736.34 |
| Email Marketing | $9,621.69 | $21,167.72 | +$11,546.03 | +120.0% | $217,266.52 | $56,179.16 |
| Google Ads | $41,426.44 | $16,570.58 | $-24,855.86 | -60.0% | $43,751.81 | $-1,563.71 |
| Amazon Sponsored Display | $38,684.39 | $15,473.76 | $-23,210.63 | -60.0% | $24,361.07 | $-7,191.00 |
| Meta Ads | $25,035.57 | $10,014.23 | $-15,021.34 | -60.0% | $12,256.87 | $-5,540.47 |

## 5. Prioritized Actionable Recommendations

| Priority | Target | Category | Action & Rationale | Estimated Impact |
|---|---|---|---|---|
| <ParagraphStyle 'BadgeHigh'> | Amazon Sponsored Products | Budget Reallocation | Gradually scale daily budget by $37,186.69 (+44.8%) across high-performing campaigns, monitoring weekly marginal ROAS to prevent premature saturation. Current spend: $83,025.48, Recommended spend: $120,212.17 (+44.8%), Expected ROAS: 3.58x, Marginal ROAS: 2.52x | +$37,186.69 reallocation into high-efficiency channel. |
| <ParagraphStyle 'BadgeHigh'> | Affiliate Marketing | Budget Reallocation | Gradually scale daily budget by $14,673.69 (+93.0%) across high-performing campaigns, monitoring weekly marginal ROAS to prevent premature saturation. Current spend: $15,771.33, Recommended spend: $30,445.02 (+93.0%), Expected ROAS: 4.39x, Marginal ROAS: 2.88x | +$14,673.69 reallocation into high-efficiency channel. |
| <ParagraphStyle 'BadgeHigh'> | Email Marketing | Budget Reallocation | Gradually scale daily budget by $11,546.03 (+120.0%) across high-performing campaigns, monitoring weekly marginal ROAS to prevent premature saturation. Current spend: $9,621.69, Recommended spend: $21,167.72 (+120.0%), Expected ROAS: 10.26x, Marginal ROAS: 6.50x | +$11,546.03 reallocation into high-efficiency channel. |
| <ParagraphStyle 'BadgeHigh'> | Google Ads | Budget Reallocation | Reduce channel allocation by $24,855.86 (-60.0%), pruning unproductive target keywords and ad sets before deploying remaining funds. Current spend: $41,426.44, Recommended: $16,570.58 (-60.0%), Historical ROAS: 2.64x vs Target 2.50x | Saves $24,855.86 in inefficient ad expenditure. |
| <ParagraphStyle 'BadgeHigh'> | Amazon Sponsored Display | Budget Reallocation | Reduce channel allocation by $23,210.63 (-60.0%), pruning unproductive target keywords and ad sets before deploying remaining funds. Current spend: $38,684.39, Recommended: $15,473.76 (-60.0%), Historical ROAS: 1.57x vs Target 2.50x | Saves $23,210.63 in inefficient ad expenditure. |
| <ParagraphStyle 'BadgeHigh'> | Meta Ads | Budget Reallocation | Reduce channel allocation by $15,021.34 (-60.0%), pruning unproductive target keywords and ad sets before deploying remaining funds. Current spend: $25,035.57, Recommended: $10,014.23 (-60.0%), Historical ROAS: 1.22x vs Target 2.50x | Saves $15,021.34 in inefficient ad expenditure. |

**GOVERNANCE & HUMAN APPROVAL NOTICE:**
This report provides automated decision-support recommendations generated from machine learning models and constrained mathematical optimization. It does not automatically modify active Amazon Ads or external ad network campaigns. All budget shifts and bid changes require explicit review and authorization by authorized marketing leadership before live execution.
**Disclaimer:** Independent demonstration project utilizing synthetic e-commerce data. Not an official Amazon application.